CHAPTER 3

FOURIER SERIES

In the preceding chapter, we have exploited the idea of resolving a signal into a weighted sum of a chosen set of basis signals as

$$x(t) = \sum_{k=-\infty}^{\infty} a_k \varphi_k(t) \tag{3.1}$$

with the intention that if the response of a **LTI** system to the signals are known, the response to $x(t)$ can be expressed as a weighted sum of these responses. In Chapter 2, $\varphi_k(t)$ were chosen to be advanced and delayed versions of $\varphi_0(t)$, which was a good choice in many respects.

Another possibility is to select the basis functions such that the response to $\varphi_k(t)$ is $\lambda_k \varphi_k(t)$. That is $\varphi_k(t)$can be selected as the <u>eigen functions</u> of **LTI** systems, with the corresponding <u>eigenvalues</u> λ_k.

If such a choice is made, then for the input signal in (3.1), the corresponding output signal will be

$$y(t) = \sum_{k=-\infty}^{\infty} a_k \lambda_k \varphi_k(t) \tag{3.2}$$

As a result, the input-output relationship of a system can be specified by its eigenvalues λ_k, k=...-1,0,1,2,...

3.1. THE RESPONSE OF LTI SYSTEMS TO COMPLEX EXPONENTIALS

For **LTI** systems of practical interest, the eigenfunctions are in the form of complex exponentials

$$\varphi_k(\mathrm{t}) = e^{s_k t}$$

for some complex number $\boldsymbol{s_k}$. Let $y(t)$ denote the response of **LTI** system to the input e^{st}, i.e.

$$e^{st} \rightarrow y(t). \tag{3.1.1}$$

Because of time invariance the response to a shifted complex exponential is

$$e^{s(t+\tau)} \rightarrow y(t+\tau) \tag{3.1.2}$$

where τ is a fixed number.

On the other hand, since the input can be expressed as $e^{st}e^{s\tau}$ and since the system is linear, (3.1.1) implies

$$e^{st}e^{s\tau} \rightarrow e^{st}y(t) \tag{3.1.3}$$

Combining (3.1.2) and (3.1.3)

$$y(t+\tau) = e^{st}y(t). \tag{3.1.4}$$

Setting $t = 0$ in (3.1.4) one obtains

$$y(\tau) = e^{st}y(0) \tag{3.1.5}$$

for all $\boldsymbol{\tau}$. Renaming $\boldsymbol{\tau}$ as $\boldsymbol{t}$, one arrives at the conclusion

$$e^{st} \rightarrow y(0)e^{st} \tag{3.1.6}$$

and thus e^{st} is an <u>eigenfunction</u> with the <u>eigenvalue</u> $y(0)$, provided $y(0)$ is finite.

Next questions is to determine the conditions under which $y(0)$ is finite.

If $h(t)$ is the unit impulse response of the system in question,

$$y(\tau) = e^{st} * h(t) = \int_{-\infty}^{\infty} e^{s(t-\tau)}h(\tau)d\tau \tag{3.1.7}$$

and thus

$$y(0) = \int_{-\infty}^{\infty} e^{s\tau}h(\tau)d\tau \ . \tag{3.1.8}$$

Hence e^{st} is an eigenfunction of a certain **LTI** system whenever

$$H(s) = \int_{-\infty}^{\infty} e^{s\tau}h(\tau)d\tau \ . \tag{3.1.9}$$

is finite and $H(s)$ is the corresponding eigenvalue. We will return to (3.1.9) later to investigate convergence properties of Fourier series.

3.2. LINEAR COMBINATIONS OF HARMONICALLY RELATED EXPONENTIALS

As we have mentioned in Chapter 1, a periodic signal is defined as one which satisfies

$$x(t) = x(t+T) \tag{3.2.1}$$

for all $\boldsymbol{t}$ and some positive non-zero T. The <u>fundamental period</u> $\boldsymbol{T_0}$ was defined as the minimum non-zero $\boldsymbol{T}$ for which (3.2.1) holds and the value $\boldsymbol{2\pi/T_0}$ was referred to as the <u>fundamental frequency.</u>

Two basic periodic signals are the sinusoid

$$x(t) = \cos\omega_0 t \tag{3.2.2}$$

and the periodic complex exponential

$$x(t) = e^{j\omega_0 t} \tag{3.2.3}$$

both with fundamental frequency ω_0 and fundamental period $T_0 = 2\pi/\omega_0$.

The set of harmonically related complex exponentials

$$\varphi_k(t) = e^{jk\omega_0 t}\ ; \quad k=0, \mp 1, \mp 2, \ldots\ldots \tag{3.2.4}$$

are derived from (3.2.3). Each of these signals is periodic with a fundamental frequency that is an integer multiple of ω_0 . Therefore they all have a common <u>period</u> $\boldsymbol{T_0}$ As a result of this, a linear combination of harmonically related exponentials

$$x(t) = \sum_{k=-\infty}^{\infty} a_k\, e^{jk\omega_0 t} \tag{3.2.5}$$

is also periodic with period $\boldsymbol{T_0}$. In (3.2.5) the terms for k=l and k=-l both have fundamental periods equal to $\boldsymbol{T_0}$ and are called the <u>fundamental components.</u> The term for k=0 is a dc or constant term, the two terms for k=2 and k=-2 are the <u>second harmonic components,</u> and more generally the terms for k=N and k=-N are the <u>N'th harmonic components.</u>

A representation in the form (3.2.5) is called the <u>Fourier series</u> representation.

<u>Example 3.2.1</u>

Consider the periodic signal

$$x(t) = \sum_{k=-3}^{3} a_k\, e^{jkt}$$

with fundamental frequency 1, where a_0=1, a_1=a_{-1}=3/4, a_2=a_{-2}=-l/2 and a_3=a_{-3}=l/4. Rewriting and combining terms using Euler's relation

$$x(t) = 1 + \frac{3}{4}\left(e^{jt} + e^{-jt}\right) - \frac{1}{2}\left(e^{j2t} + e^{-j2t}\right) + \frac{1}{4}\left(e^{j3t} + e^{-j3t}\right)$$
$$= 1 + \frac{3}{2}\cos t - \cos 2t + \frac{1}{2}\cos 3t$$

Fig.3.2.1. illustrates how this signal is built up from its harmonic components.

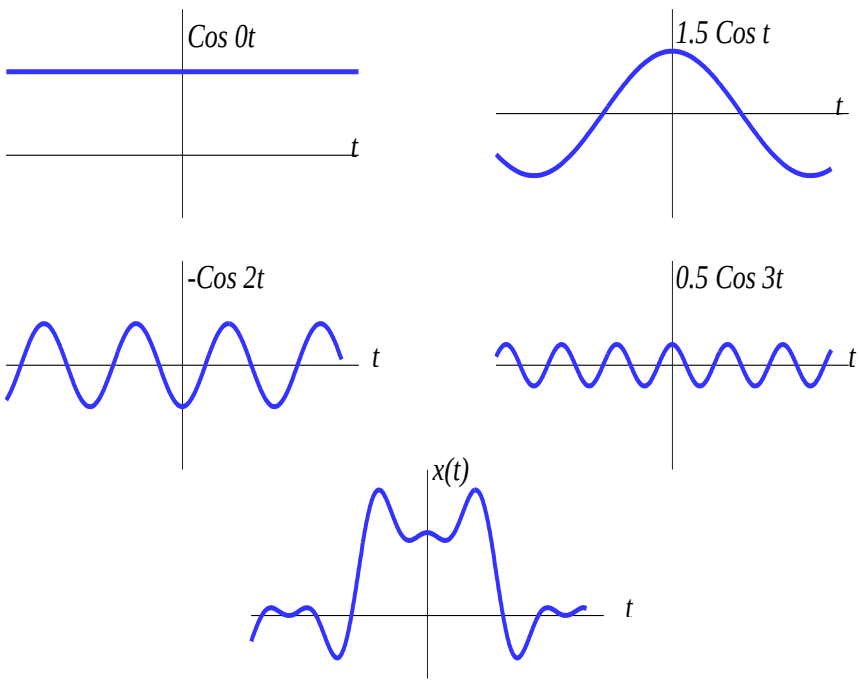

Fig. 3.2.1. Example 3.2.1

If $x(t)$ is a real periodic function which can be expressed in the form (3.2.5), then since

$$x^*(t) = \sum_{k=-\infty}^{\infty} a_k^* \, e^{-jk\omega_0 t} \tag{3.2.6}$$

one has

$$\begin{aligned} x^*(t) &= \sum_{k=-\infty}^{\infty} a_k^* \, e^{-jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_{-k}^* \, e^{jk\omega_0 t} \\ &= x(t) = \sum_{k=-\infty}^{\infty} a_k \, e^{jk\omega_0 t} \end{aligned} \tag{3.2.7}$$

Therefore one has the relation

$$a_k^* = a_{-k} \tag{3.2.8}$$

for real periodic signals.

Alternative forms of Fourier series can be derived by rearranging the terms in (3.2.5) as

$$x(t) = a_0 + \sum_{k=1}^{\infty} \{a_k e^{jk\omega_0 t} + a_{-k} e^{-jk\omega_0 t}\} \tag{3.2.9}$$

When the signal $x(t)$ in question is real, using (3.2.8), (3.2.9) becomes

$$\begin{aligned} x(t) &= a_0 + \sum_{k=1}^{\infty} \{a_k e^{jk\omega_0 t} + a_k^* e^{-jk\omega_0 t}\} \\ &= a_0 + \sum_{k=1}^{\infty} 2\mathcal{R}e\{a_k e^{jk\omega_0 t}\} \end{aligned} \tag{3.2.10}$$

and when the complex coefficients $\boldsymbol{a_k}$ are expressed in polar form

$$a_k = A_k e^{j\theta_k} \tag{3.2.11}$$

(3.2.10) becomes

$$x(t) = a_0 + 2\sum_{k=1}^{\infty} A_k \cos(k\omega_0 t + \theta_k) \tag{3.2.12}$$

As can easily be verified by the reader, another possible representation is

$$x(t) = a_0 + 2\sum_{k=1}^{\infty} \{B_k \cos(k\omega_0 t) - C_k \sin(k\omega_0 t)\} \tag{3.2.13}$$

where

$$a_k = B_k + jC_k. \tag{3.2.14}$$

Although the two forms given by (3.2.12) and (3.2.14) are more common, the complex exponential form given by (3.2.5) is more convenient for our purposes and will be used exclusively.

In general when a periodic signal is applied at the input of a **LTI** system, the output is also periodic with the same period. Let $x(t)$ be a periodic signal with a Fourier series representation (3.2.5) and suppose that we apply this signal to a **LTI** system with the impulse response $h(t)$. Using the fact that the complex exponentials are eigenfunctions of the system, it follows from (3.1.6-9) that

$$y(t) = \sum_{k=-\infty}^{\infty} a_k H(jk\omega_0) \, e^{jk\omega_0 t} \tag{3.2.15}$$

where

$$H(jk\omega_0) = \int_{-\infty}^{\infty} h(\tau) e^{-jk\omega_0 \tau}\, d\tau \quad . \tag{3.2.16}$$

Example 3.2.2

Let the periodic signal x(t) discussed in Example 3.2.1 be the input to a LTI system with the impulse response

$h(t) = e^{-t} u(t)$

To calculate the output $y(t)$*, let us first compute* $H(jk\omega_0)$*:*

$$H(jk\omega_0) = \int_0^{\infty} e^{-\tau} e^{-jk\tau}\, d\tau$$

$$= \frac{-1}{1+jk} e^{-(1+jk)\tau}\Big|_0^{\infty} = \frac{1}{1+jk}$$

Therefore from (3.2.15)

$$y(t) = \sum_{k=-3}^{3} b_k e^{jkt}$$

with $b_k = a_k H(jk\omega_0)$*, i.e.*

$$b_0 = 1$$

$$b_1 = \frac{3}{4}\frac{1}{1+j} \qquad b_{-1} = \frac{3}{4}\frac{1}{1-j}$$

$$b_2 = \frac{-1}{2}\frac{1}{1+2j} \qquad b_{-2} = \frac{-1}{2}\frac{1}{1-2j} \quad .$$

$$b_3 = \frac{1}{4}\frac{1}{1+3j} \qquad b_{-3} = \frac{1}{4}\frac{1}{1-3j}$$

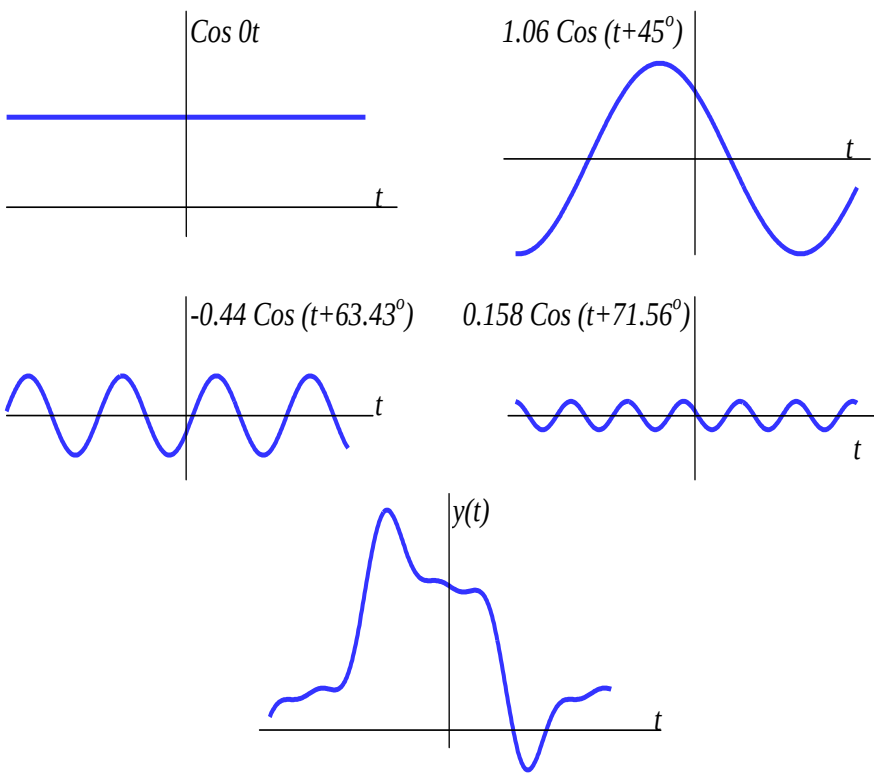

Fig. 3.2.2. Example 3.2.2

3.3. DETERMINATION OF THE FOURIER SERIES

Assuming that a given periodic signal x(t) has a Fourier series representation, we wish to express the coefficients a_k of the Fourier series expansion

$$x(t) = \sum_{k=-\infty}^{\infty} a_k\, e^{jk\omega_0 t} \tag{3.3.1}$$

Multiplying both sides by $e^{jn\omega_0 t}$ and integrating over one period (from 0 to $T_0 = 2\pi/\omega_0$, we have

$$\begin{aligned}\int_0^{T_0} e^{-jn\omega_0 t}dt &= \int_0^{T_0} \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0} e^{-jn\omega_0 t}dt \\ &= \sum_{k=-\infty}^{\infty} a_k \left[\int_0^{T_0} e^{j(k-n)t}dt\right]\end{aligned} \tag{3.3.2}$$

Rewriting the integral in square brackets and using the Euler's identity,

$$\begin{aligned}\int_0^{T_0} e^{-j(k-n)\omega_0 t}dt &= \int_0^{T_0} \{\cos(k-n)\omega_0 t + j\sin(k-n)\omega_0 t\}dt \\ &= \begin{cases} 0 & n \neq k \\ T_0 & n = k \end{cases}\end{aligned} \tag{3.3.4}$$

and consequently the right hand side of (3.3.2) is equal to $a_n T_0$. Therefore,

$$a_n = \frac{1}{T_0}\int_0^{T_0} x(t)e^{-jn\omega_0 t}dt\ . \tag{3.3.5}$$

Note that the integral in (3.3.5) can be taken over <u>any interval of length T_0</u>, since (3.3.4) still holds if the limits of integration are t_0 and t_0+T_0 for any t_0.

To summarize, if $x(t)$ has a Fourier series representation then the coefficients are given by (3.3.5). (3.2.5) is often referred to as the <u>synthesis</u> equation and (3.3.5) as the <u>analysis</u> equation. The coefficients $\{a_k\}$ are often called the <u>Fourier series coefficients</u> or the <u>spectral coefficients</u> of $x(t)$. Let us mention once again that the totality of the spectral coefficients $\{a_k\}$ completely characterize a <u>periodic</u> signal.

<u>Example 3.3.1</u>

Suppose that $x(t)$ is a sawtooth wave as shown in Fig.3.3.1.

This signal is defined to be equal to t/T in the interval $(T/2, -T/2,)$. Hence

$$a_n = \frac{1}{T}\int_{-T/2}^{T/2} \frac{t}{T}e^{-jn\omega_0 t}dt = \frac{j}{2\pi n}(-1)^n \quad ; \quad n\neq 0.$$

and

$$a_0 = \frac{1}{T}\int_{-T/2}^{T/2} \frac{t}{T}dt = \frac{1}{T^2}\frac{t^2}{2}\Big|_{-T/2}^{T/2} = 0$$

Therefore ,

$$x(t) = \ldots\ldots + \frac{j}{4\pi}e^{-2j\omega_0 t} - \frac{j}{2\pi}e^{-j\omega_0 t} + \frac{j}{2\pi}e^{j\omega_0 t} - \frac{j}{4\pi}e^{2j\omega_0 t} + \frac{j}{6\pi}e^{3j\omega_0 t}$$
$$= \frac{1}{\pi}\left\{\sin\omega_0 t - \frac{1}{2}\sin 2\omega_0 t + \frac{1}{3}\sin 3\omega_0 t + \ldots\ldots\right\}$$
$$= \frac{1}{\pi}\sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k}\sin k\omega_0 t$$

where $\omega_0 = 2\pi/T$.

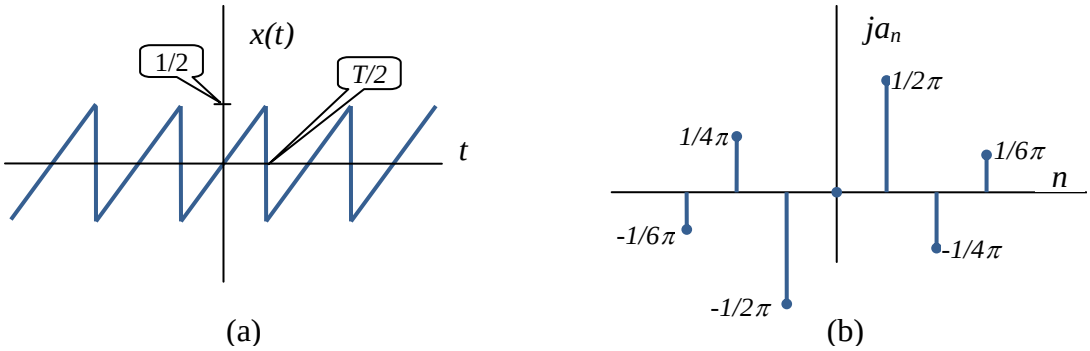

Fig.3.3.1. Example 3.3.1

Example 3.3.2

Consider the half-wave rectified sine wave

$$x(t) = \begin{cases} \sin\pi t & 2k \le t < 2k+1 \\ 0 & 2k-1 \le t < 2k \end{cases}$$

which is depicted in Fig.3.3.2.

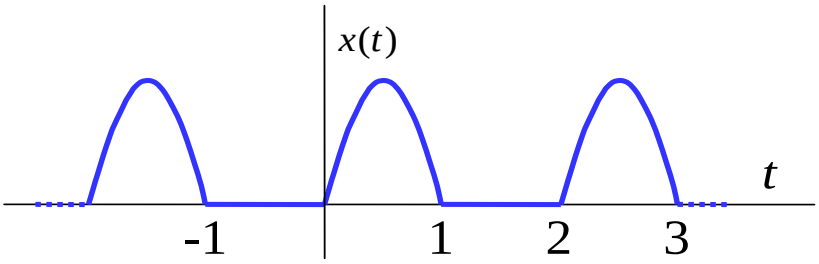

Fig. 3.3.2. Example 3.3.2

The spectral coefficients of $x(t)$ *can be computed as*

$$a_n = \frac{1}{2}\int_0^2 x(t)e^{jn\omega_0 t}dt = \frac{1}{2}\int_0^1 \sin\pi t.\, e^{jn\omega_0 t}dt = \frac{1}{4j}\int_0^1 \left[e^{-j\pi(n-1)t} - e^{-j\pi(n+1)t}\right]$$
$$= \begin{cases} \dfrac{1}{\pi(1-n^2)} & \text{n=even} \\ -j/4 & \text{n=1} \\ j/4 & \text{n=-1} \\ 0 & \text{otherwise} \end{cases}$$

Thus, we have

$$x(t)=\begin{cases}\frac{1}{\pi}-\frac{j}{4}e^{-j\pi t}+\frac{j}{4}e^{j\pi t}+\sum_{k=1}^{\infty}\frac{1}{\pi[1-(2k)^2]}\left(e^{j2k\pi t}+e^{-j2k\pi t}\right)\\ \frac{1}{\pi}+\frac{1}{2}\sin\pi t+\sum_{k=1}^{\infty}\frac{2}{\pi(1-4k^2)}\cos 2k\pi t\end{cases}.$$

3.4. EXISTENCE AND CONVERGENCE OF FOURIER SERIES

The reasoning in the preceeding sections leads to the conclusion that if a function $x(t)$ equals the sum in (3.2.5) then the coefficients of this sum are given by (3.3.5). This is not of course, a proof that any periodic function can be expanded into a series as in (3.2.5). To prove the existence of such an expansion one must show that if the coefficients are calculated from (3.3.5), and the results are inserted in (3.2.5), the sum equals the function $x(t)$.

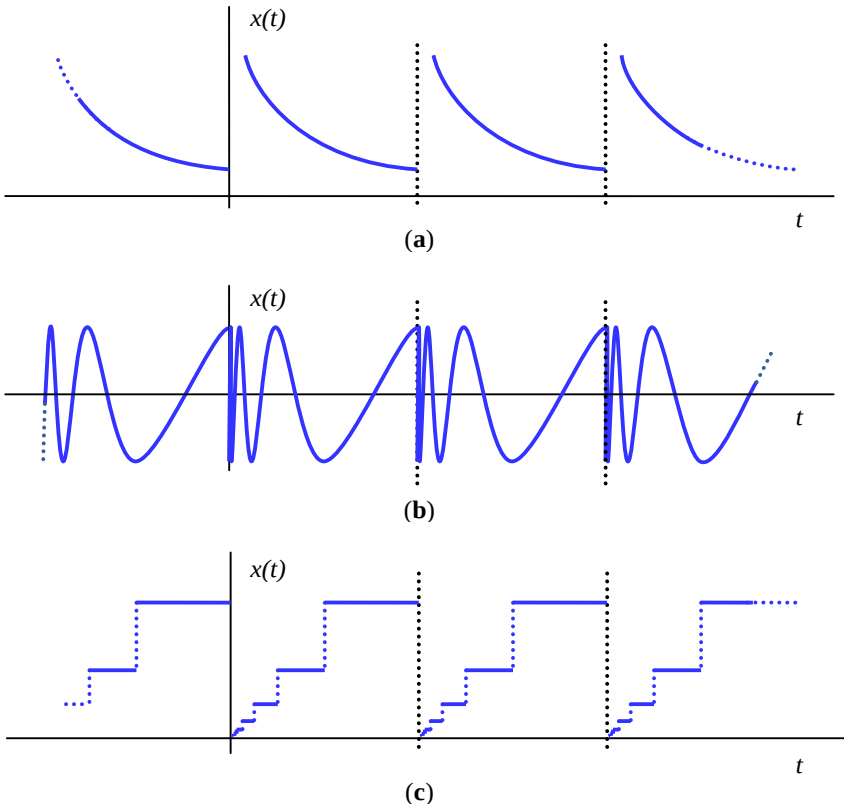

Fig.3.4.1. Signals that violate Dirichlet conditions:

a) $x(t)=\frac{1}{t-kT}$ $kT<t\leq(k+1)T$ violating first condition;

b) $x(t)=\cos\left(\frac{1}{t-kT}\right)$ $kT<t\leq(k+1)T$ violating second condition;

c) $x(t)=\frac{1}{2^i}$ $\frac{T}{i+1}<t-kT\leq\frac{T}{i}$ i=1. 2. 3. ..... for $kT<t\leq(k+1)T$ violating third condition

As a matter of fact, the decomposition in (3.2.5) is possible if $x(t)$ is a periodic function satisfying all of the following conditions:

i) $x(t)$ is absolutely integrable, i.e.

$$\int_0^T|x(t)|dt<\infty\ , \tag{3.4.1}$$

ii) $x(t)$ has a finite number of maxima and minima within a period,

iii) $x(t)$ has a finite number of discontinuities within a period.

These conditions are known as the Dirichlet conditions. Examples of periodic signals violating first, second and third Dirichlet conditions are given in Fig.3.4.1 (a), (b) and (c) respectively.

Let us note that since the complex exponential function is continuous for every ***t***, the right hand side of (3.2.5) is also continuous for every ***t***. Thus an interesting question arises: What value it will have at $t=t_0$, if $x(t)$ has a discontinuity at $t=t_0$? As it turns out it can be shown that the two sides of (3.2.5) are equal at every continuity point of $x(t)$. At the discontinuities, however, the sum on the right hand side equals the average of the limits $f(t^-)$ and $f(t^+)$, that is, if $x(t)$ is discontinuous at $t=t_0$, then

$$\sum_{k=-\infty}^{\infty} a_k\, e^{jk\omega_0 t_0} = \frac{1}{2}\{x(t_0^-) + x(t_0^+)\} \ . \tag{3.4.2}$$

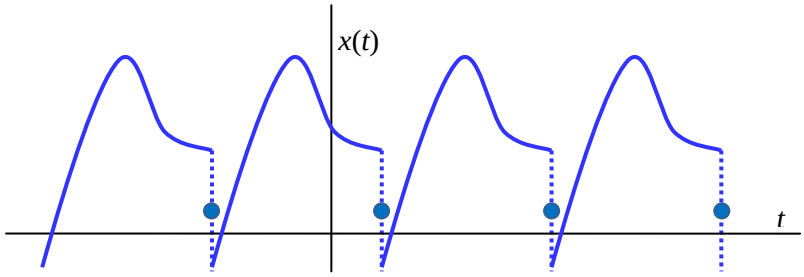

Fig. 3.4.2 A periodic signal with discontinuities

Example 3.4.1

Let us determine the value of the Fourier series expansion of the sawtooth wave of Example 3.3.1 at t=T/2 :

$$x(T/2) = \frac{1}{\pi}\sum_{k=1}^{\infty}\frac{(-1)^k}{k}\sin(k\omega_0\, T/2)$$
$$= \frac{1}{\pi}\sum_{k=1}^{\infty}\frac{(-1)^k}{k}\sin(k\pi) = 0 \ .$$

On the other hand $x(T/2^-) = 1/2$ *and* $x(T/2^+) = 1/2$ *and thus verifying the theory.*

3.5. PROPERTIES OF FOURIER SERIES

3.5.1. Even and odd Functions

If a function $x(t)$ is even, then since

$$x(t) = x(-t) = \sum_{k=-\infty}^{\infty} a_k\, e^{-jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k\, e^{jk\omega_0 t} \tag{3.5.1}$$

one has the meaningful relation

$$a_k = a_{-k} \ , \tag{3.5.2}$$

and hence

$$x(t) = a_0 + \sum_{k=1}^{\infty} a_k \left[e^{jk\omega_0 t} + e^{-jk\omega_0 t}\right]$$
$$= a_0 + 2\sum_{k=1}^{\infty} a_k \cos(k\omega_0 t) \tag{3.5.3}$$

contains only cosine terms. On the other hand if $x(t)$ is odd, then

$$a_k = -a_{-k} \quad , \qquad a_0 = 0 \tag{3.5.4}$$

and therefore

$$x(t) = 2j\sum_{k=1}^{\infty} a_k \sin(k\omega_0 t) \quad . \tag{3.5.5}$$

A function may have additional symmetries within a period. For example the function in Fig.3.5.1(a) satisfies $x\left(t + {}^{T}/_{2}\right) = -x(t)$.

Such a signal will have only odd harmonics, i.e. $a_n=0$ for n even. Proof is left as an excercise for the reader. Similarly, if $x(t)$ is periodic with $x\left(t + {}^{T}/_{2}\right) = x(t)$, the Fourier series will consist of even harmonics only. However, in this case the fundamental period is not T but $T/2$ and the fundamental frequency is $4\pi/T$ explaining the existence of even harmonics only.

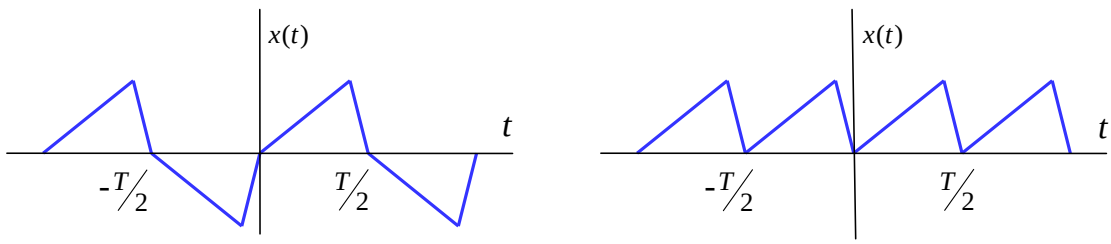

Fig.3.5.1. Periodic signals with property (a) $x(t+T/2) = -x(t)$ (b) $x(t+T/2) = x(t)$

3.5.2. Manipulation of Signals

If we know the Fourier series expansion of a function $x(t)$, we can determine the expansion of the shifted function $\tilde{x}(t) = x(t - t_0)$ simply by replacing ***t*** by t-t_0:

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} \tilde{a}_k\, e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k\, e^{jk\omega_0 t} e^{-jk\omega_0 t_0} \tag{3.5.6}$$

and thus

$$\tilde{a}_k = \sum_{k=-\infty}^{\infty} a_k\, e^{jk\omega_0 t} \tag{3.5.7}$$

If $x(t)$ is a periodic signal, so is $x(-t)$ with the same fundamental period. The Fourier coefficients of $\tilde{x}(t) = x(-t)$ can be obtained from

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} \tilde{a}_k\, e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k\, e^{-jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_{-k}\, e^{jk\omega_0 t} \tag{3.5.8}$$

as

$$\tilde{a}_k = a_{-k} \; . \tag{3.5.9}$$

When a periodic signal is differentiated, another periodic signal of the same fundamental period is obtained. In this case the coefficients of the Fourier series expansion of the derivative are obtained from

$$\tilde{x}(t) = \frac{dx(t)}{dt} = \sum_{k=-\infty}^{\infty} \frac{d}{dt}\left[a_k e^{jk\omega_0 t}\right]$$
$$= \sum_{k=-\infty}^{-1} jk\omega_0 a_k e^{jk\omega_0 t} + \sum_{k=1}^{\infty} jk\omega_0 a_k e^{jk\omega_0 t} \qquad (3.5.10)$$
$$= \sum_{-\infty}^{\infty} \tilde{a}_k e^{jk\omega_0 t}$$

and therefore

$$\tilde{a}_k = jk\omega_0 a_k \quad k \neq 0 \qquad (3.5.11a)$$

and

$$\tilde{a}_0 = 0\,. \qquad (3.5.11b)$$

BIBLIOGRAPHY

1. OPPENHEIM & WILLSKY: "Signals and Systems", Prentice-Hall, New Jersey, 1983.
2. LIU & LIU: "Linear System Analysis", McGraw-Hill, New York, 1975.
3. GABEL & ROBERTS: "Signals and Linear Systems", John Wiley & Sons, 1973 & 1980.
4. PAPOULIS:"Circuits and Systems", Holt-Saunders, 1980.

PROBLEMS

Problem 1.

Determine the fundamental frequency and the coefficients of the complex Fourier series for

$$x(t) = 3\,cos(3.3t) + 5\,sin(2.2t)\,.$$

Problem 2.

Determine the coefficients of the Fourier series expansion for the periodic signal

$$x(t) = \sum_{-\infty}^{\infty} (-1)^k \delta(t - 2k - 1) \quad .$$

Problem 3.

Determine the coefficients of the Fourier series expansion for the periodic signals

i) $$x(t) = \sum_{k=-\infty}^{\infty} \delta(t-k)$$

ii) $$x(t) = \sum_{k=-\infty}^{\infty} \delta(t-2k)$$

iii) $$x(t) = \sum_{k=-\infty}^{\infty} \delta(t-2k) + \sum_{k=-\infty}^{\infty} \delta(t-k)$$

Problem 4.

Consider a periodic continuous-time signal $x(t)$ with fundamental period T_0. Let $x(t)$be described by the spectral (Fourier series) coefficients a_k. A discrete-time periodic signal $x[n]$ is generated from $x(t)$ by sampling it every T_0/N seconds, {i.e. $x[n] = x(nT_0/N)$} where N is a positive integer. Express the discrete-time Fourier series coefficients bk of $x[n]$, in terms of a_k.

Problem 5.

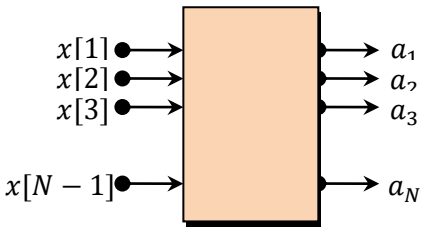

Fig. P4

The box shown in the figure represents a device or algorithm, which generates at its output, the coefficients of the discrete-time Fourier series expansion of a periodic discrete-time sequence when excited by the values of that sequence. One wishes to use this device or algorithm to determine "the periodic time-domain sequence" when the discrete-time Fourier series coefficients are known. Determine how it could be utilized for this purpose.

Problem 6.

Find the trigonometric Fourier Series for the functions,

- **a)** $f(t) = \begin{cases} 2 & 0 < t < \pi \\ 1 & \pi < t < 2\pi \end{cases}$ 2 and $f(t + 2\pi) = f(t)$;
- **b)** $f(t) = e^{-t}$ for $-\pi < t < \pi$, and $f(t + 2\pi) = f(t)$;
- **c)** $f(t) = t^2$ for $-1 < t < 1$, and $f(t + 2) = f(t)$.
- **d)** Find and sketch to scale the discrete amplitude and phase spectra of the functions of the previous parts.

Problem 7.

Find the exponential Fourier series for the signal

$$x(t) = \cos 2t + \cos t.$$

Problem 8.

The periodic signal $x(t)$ has a fundamental frequency ω_0 and spectral coefficients a_k. Express $y(t) = x(t).\cos \omega_0 t$ as a Fourrier series.

Problem 9.

Let $x[n]$ be a real-valued and odd periodic sequence with period N=7. Given that $a_{15} = j$, $a_{16} = 2j$, $a_{17} = 3j$, determine the values of a_0, a_{-1}, a_{-2}, a_{-3}, where a_k, are the Fourier series coefficients of $x[n]$.

Problem 10.

A periodic sequence $x[n]$ with fundamental period N is characterized by the spectral coefficients $\{a_k\}$. A new discrete-time sequence is defined as $y[n] = a_n$. Determine the spectral coefficients of this new sequence in terms of $x[n]$.

Problem 11.

Let $x[n]$ be a periodic sequence with period N and spectral coefficients $\{a_k\}$.

a) Express the spectral coefficients $\{b_k\}$ of the sequence $y[n] = |x[n]|^2$ in terms of $\{a_k\}$.

b) If the coefficients $\{a_k\}$ are all real, is it guaranteed that $\{b_k\}$ are also real?

Problem 12.

Two sequences $x[n]$ and $y[n]$ are periodic with the common period N. Spectral coefficients for $x[n]$ and $y[n]$ are $\{a_k\}$ and $\{b_k\}$ respectively. Determine the spectral coefficients of the following in terms of $\{a_k\}$ and $\{b_k\}$:

i) $w[n] = \begin{cases} x[n/2] & \text{for } n \text{ even} \\ 0 & \text{for } n \text{ odd} \end{cases}$.

ii) $z[n] = \begin{cases} x[n/2] & \text{for } n \text{ even} \\ y[n + 1/2] & \text{for } n \text{ odd} \end{cases}$.

iii) Repeat (ii) for the case where $y[n]$ is periodic with fundamental period M and the two periods N and M are relatively prime (i.e. they don't have a common divisor).

Problem 13.

Consider the periodic function $f(t)$ with fundamental period T_0 and Fourier series spectral coefficients $\{a_k\}$. Determine the complex Fourier series expansion of $g(t) = f(t/2) + f(2t + 1)$.

Problem 14.

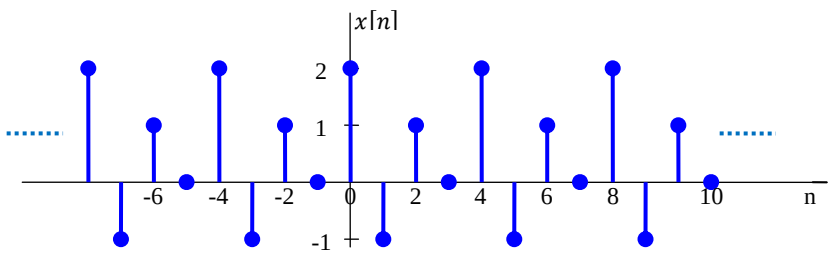

Fig. P14

Find and sketch to scale, the frequency spectrum (both magnitude and phase) of the periodic discrete-time sequence shown in the figure.

Problem 15.

Determine and sketch the output $y[n]$ of the discrete-time filter, the frequency response of which is shown in the figure, for the input

$$x[n] = (-1)^n .$$

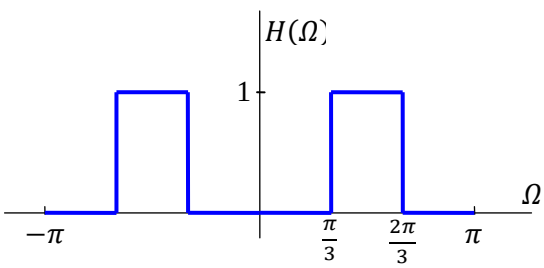

Fig. P15

Problem 16.

If $x[n]$ is a periodic sequence with period N, it is also periodic with $2N$. Let $X_1[k]$ denote the k'th Discrete Fourier series coefficient of $x[n]$ considered as a periodic sequence with period N, and $X_2[k]$ denote the same for $x[n]$ considered as a periodic sequence with period 2N. Determine $X_2[k]$ in terms of $X_1[k]$.

Problem 17.

Consider two periodic sequences $x[n]$ and $y[n]$ with periods M and N respectively. A new sequence $w[n]$ is defined as $w[n] = x[n] + y[n]$.

a) Show that $w[n]$ is periodic with period MN.

b) Since x, y and w are periodic with periods M, N and MN, their spectra are also periodic with the same periods. Determine the spectrum of $w[n]$ in terms of the spectra of $x[n]$ and $y[n]$.

Problem 18.

Let $x[n]$ and $y[n]$ be two periodic sequences with common period N and spectral coefficients a_k and b_k respectively.

Determine the exponential Fourier expansion for $x[n]y[n]$ in terms of a_k and b_k.

Problem 19.

Consider the system shown in the figure, where

$$x(t) = \frac{1}{t}\sin\frac{t}{2},$$

$$p(t) = \cos 2t + \cos 8t$$

and

$$h(t) = \sum_{k=-\infty}^{\infty} c_k e^{jkt}.$$

Determine the Fourier series representation for y(t).

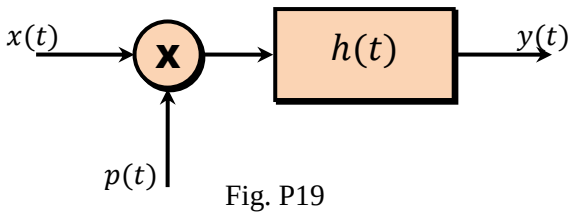

Fig. P19

Problem 20.

The r.m.s. value of a periodic signal $x(t)$ with fundamental period T_o is defined as:

$$x_{rms} = \left(\frac{1}{T_0}\int_{T}^{T+T_0} |x(t)|^2 dt\right)^{\frac{1}{2}}$$

Express x_{rms} in terms of the spectral (Fourier series) coefficients a_k of $x(t)$.

Problem 21.

Let $x[n]$ denote a periodic sequence with period N and $X[k]$ denote its k'th discrete-time Fourier series coefficient. Considering that $X[k]$ is also a periodic sequence with period N, determine the discrete-time Fourier series coefficients of $X[k]$ in terms of $x[n]$.